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Research

AI at 70 (Part 2): 14 more lessons from a lifetime of boom and bust

Context is all you need

September 2026

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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

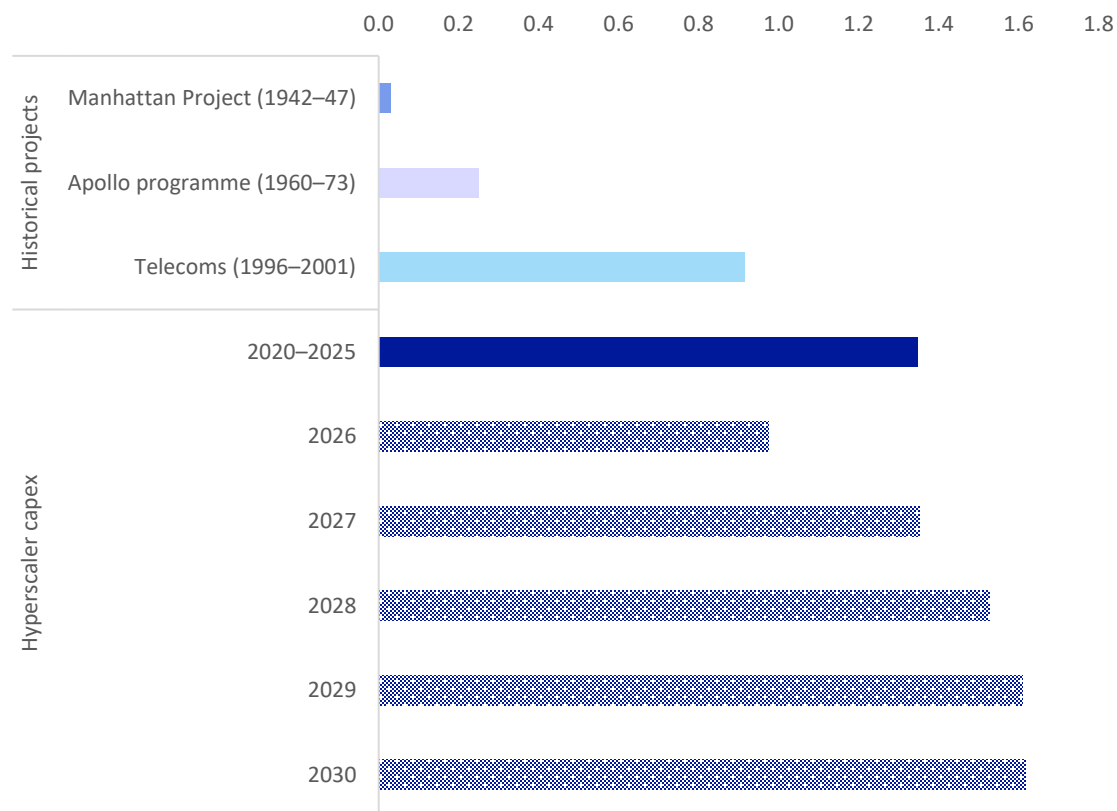
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1. This boom is remarkable not just for scale but also speed

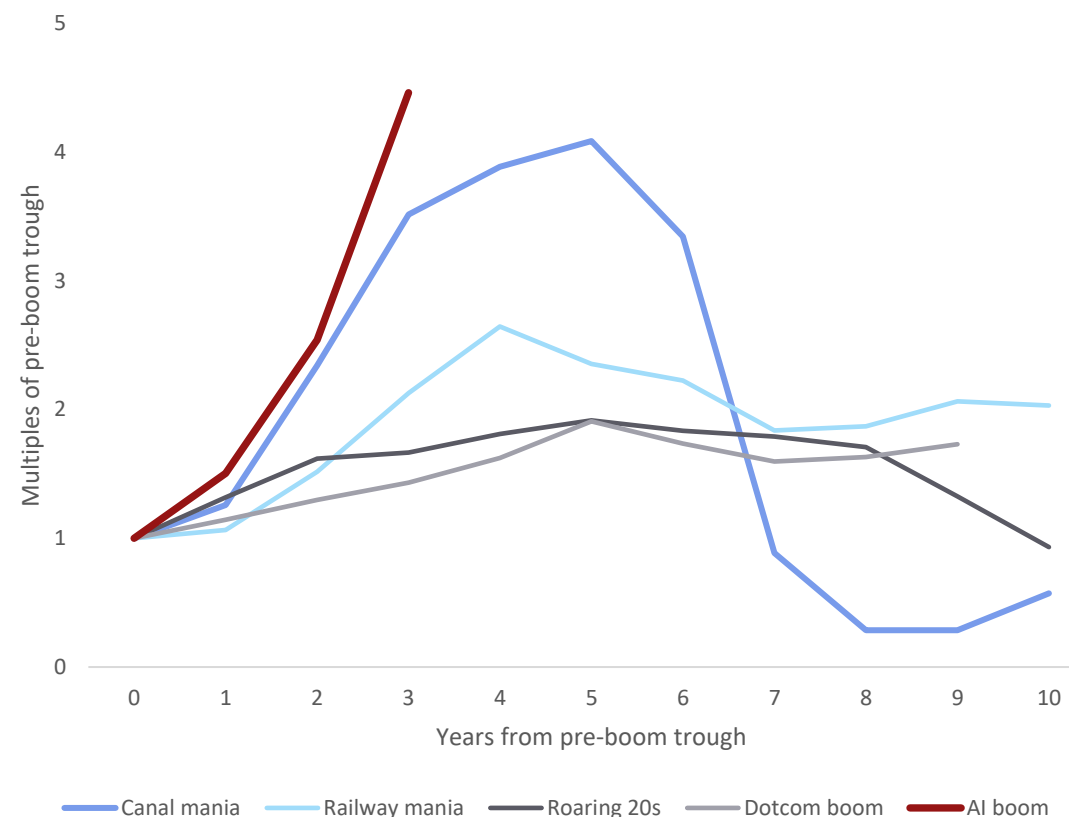
Each year of expected capex through 2030 exceeds the whole dot-com telecoms boom



Hyperscaler current and forecast capex vs major historical projects (2026 USD trn)



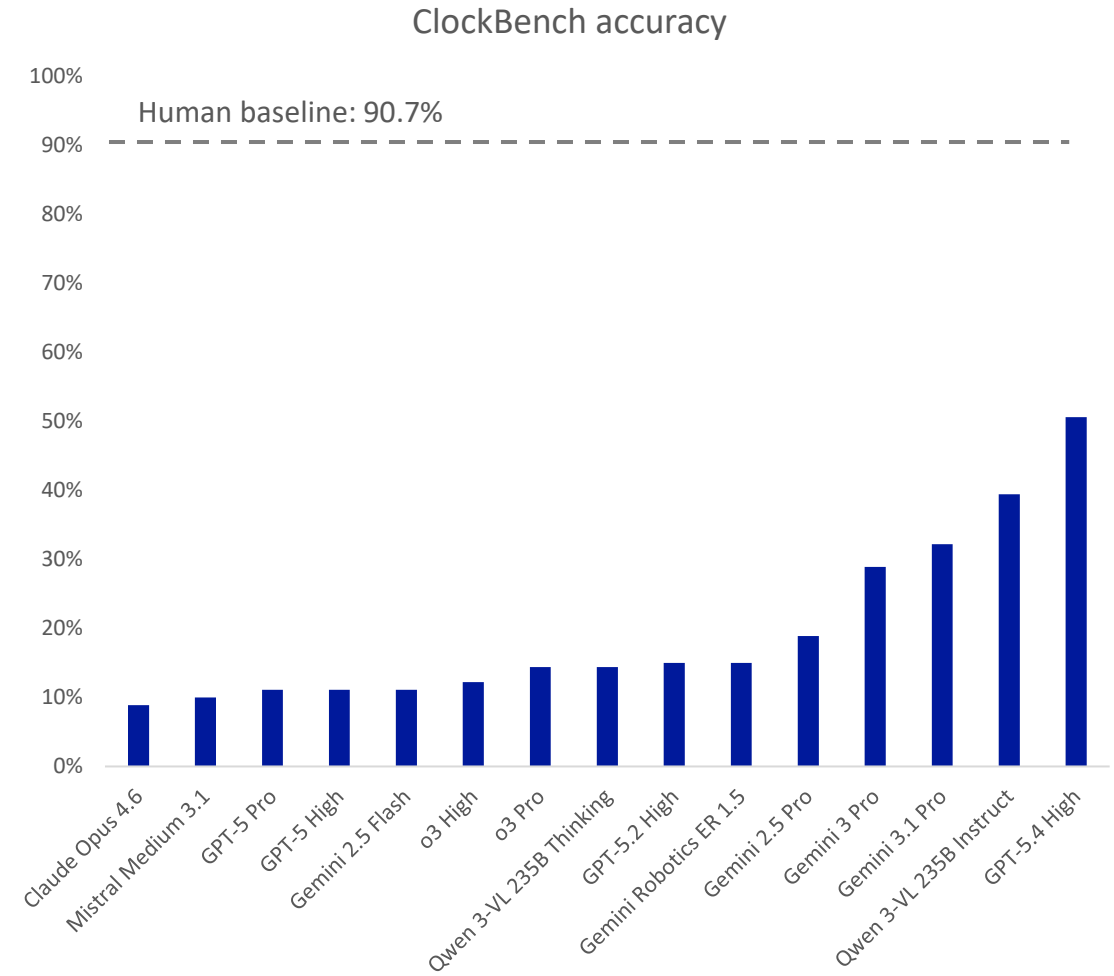
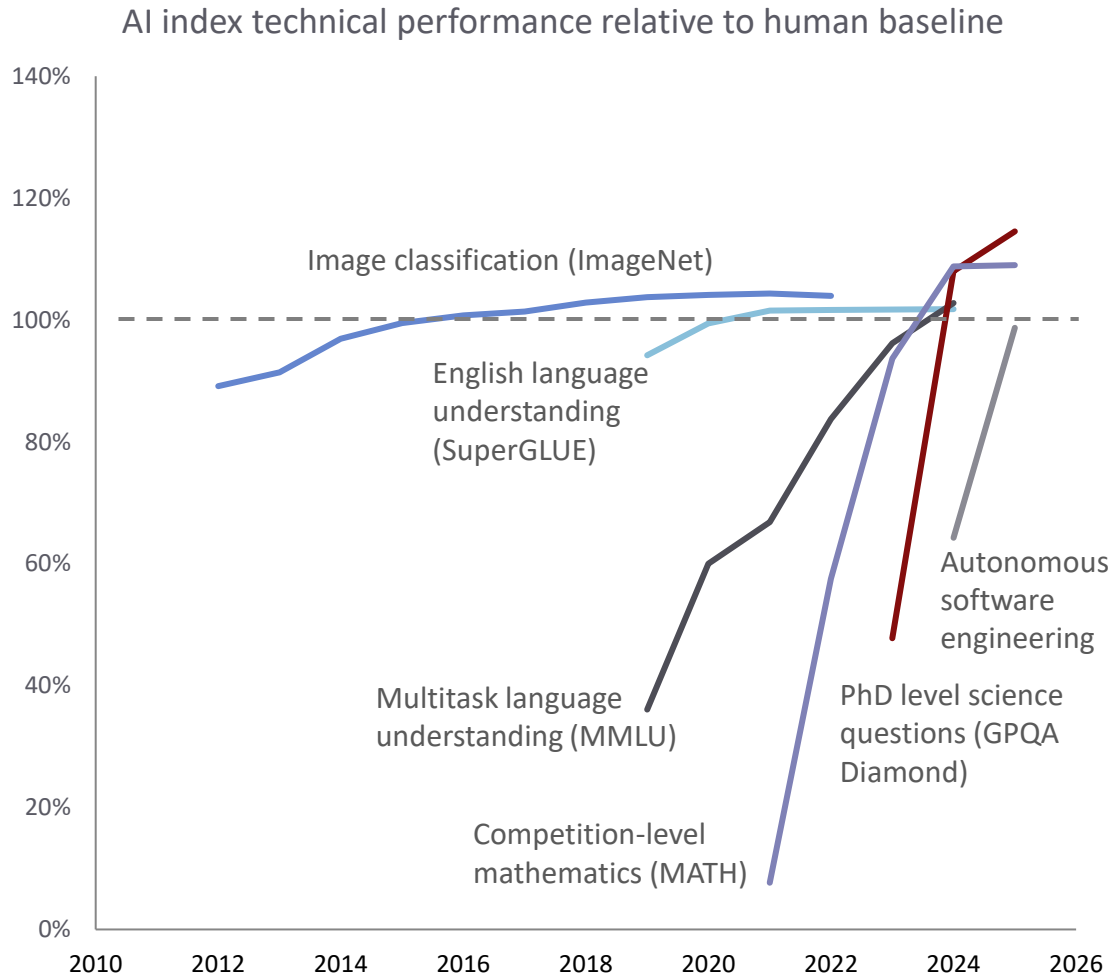
Investment during innovation booms: multiples of pre-boom trough



Source: US National Park Service, Space Policy, Brookings Institution, Bloomberg Finance LP, Bank for International Settlements, Deutsche Bank Research. Note: Hyperscaler capex estimates from Bloomberg include five US hyperscalers (AWS, Google, Meta, Microsoft and Oracle), which account for roughly 80% of the total on average, as well as tier-two/neoclouds and Chinese hyperscalers. Hyperscaler capex figures are not inflation-adjusted.

2. AI makes hard things look easy, and easy things look hard

“Jagged frontier” means AI can solve complex tasks but struggles to tell the time on a clock



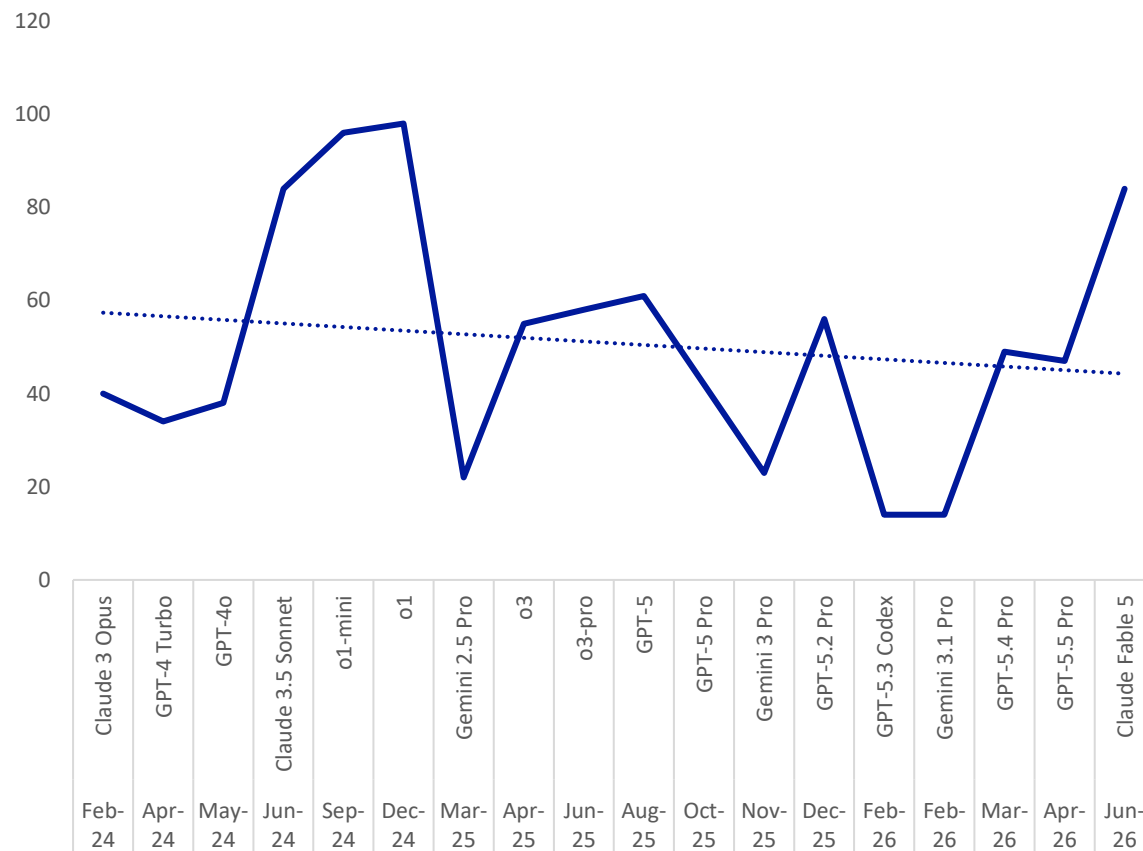
Source: Stanford HAI AI Index 2026, ClockBench Leaderboard 2026, Deutsche Bank Research

3. AI models are commoditising fast

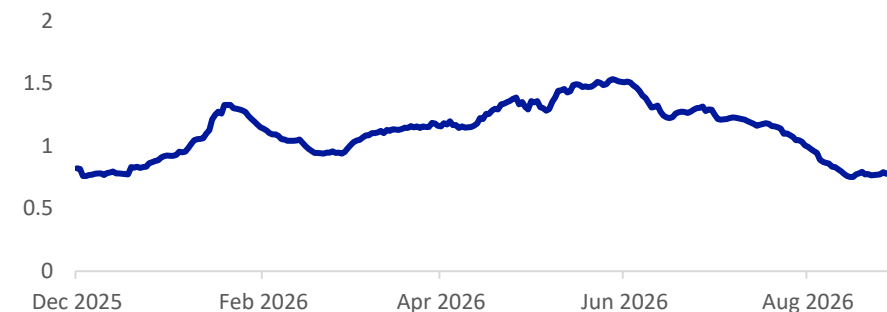
Top models hold the crown for less than two months on average and pricing is under pressure



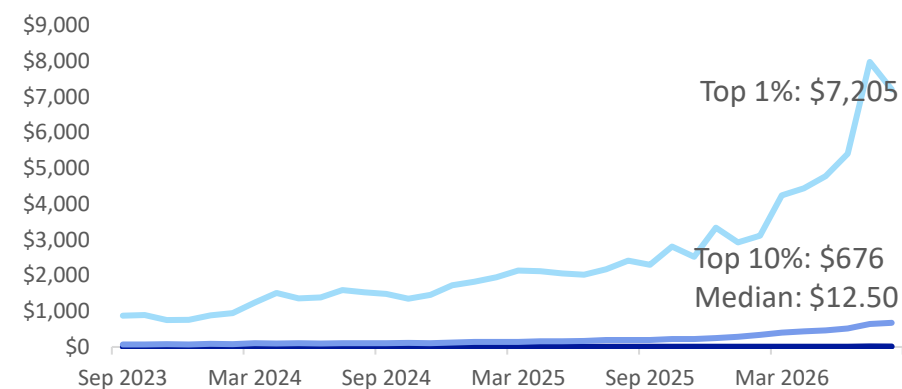
Epoch Capabilities Index: leading model tenure at the top (days)



Silicon Data LLM Token Expenditure Index (since Dec 2025)



Ramp AI Index monthly spend per employee



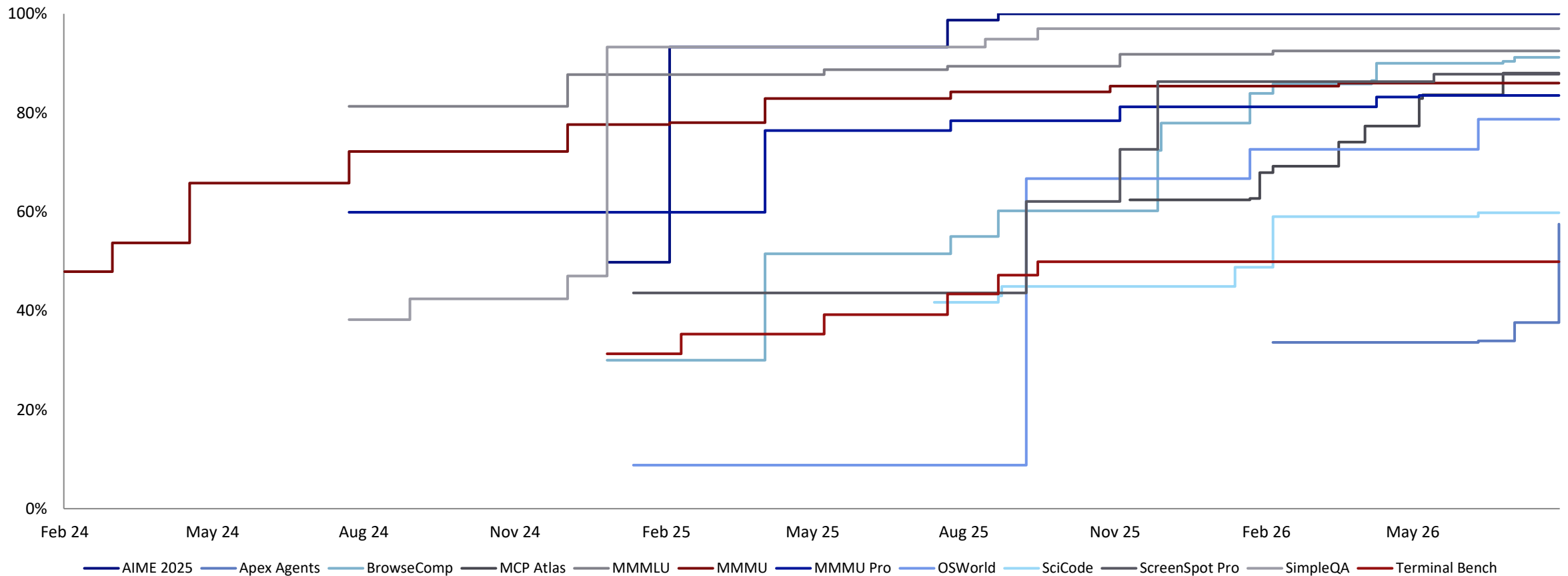
Source: Epoch, Bloomberg Finance LP, Silicon Data, Ramp AI Index, Deutsche Bank Research. NB: For clarity, Epoch Capabilities Index chart excludes OpenAI's GPT-4, the model that predated Claude 3 Opus and was at the top for 352 days, as well as the new leader since Sept 3, OpenAI's GPT-6 Astra

4. Benchmarks are ageing even faster than models

Plateaus may mean models have stopped improving... or the test is no longer hard enough



Benchmark saturation: State of the Art (SOTA) score progression over time -- each step is a new best model



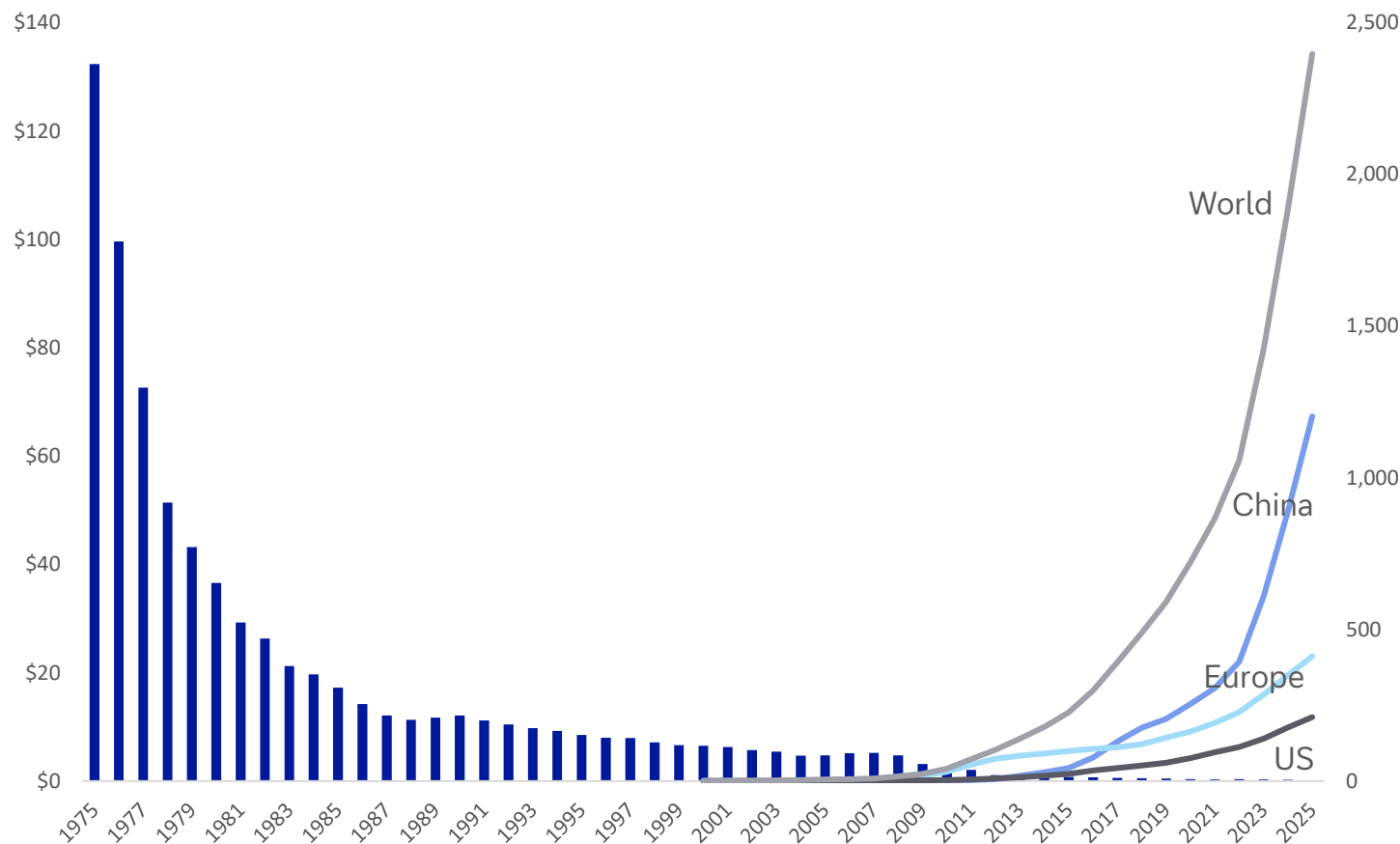
Source: [LLM Stats](#), Deutsche Bank Research

5. Tech use accelerates when the cost falls far enough

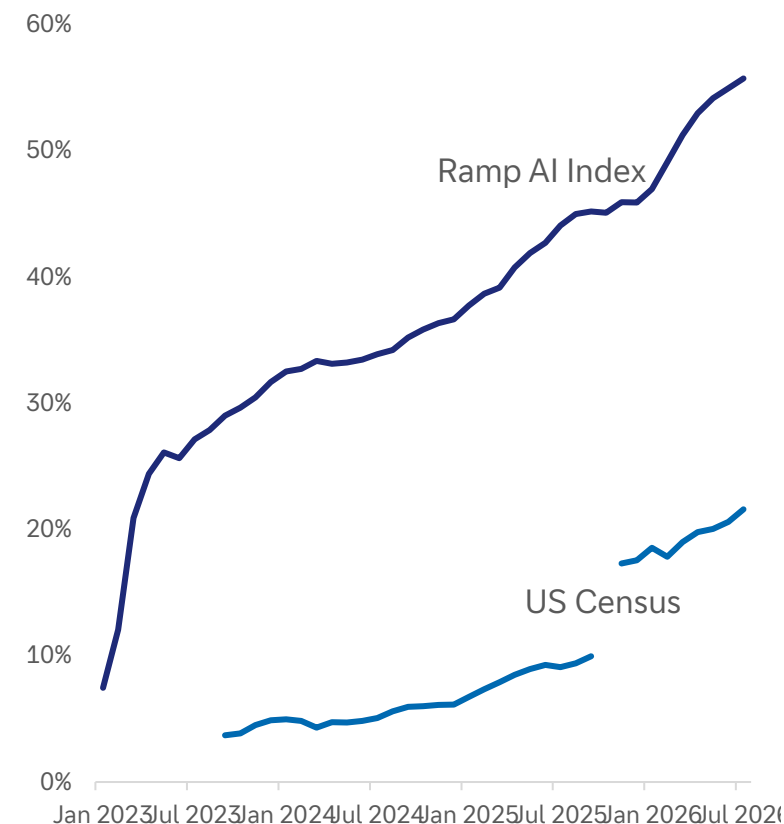
Solar and now AI are following similar patterns to horsepower, electricity and mainframes



Solar voltaic panel prices (USD/watt, adjusted for inflation, lhs) vs installed solar energy capacity (GW, rhs)



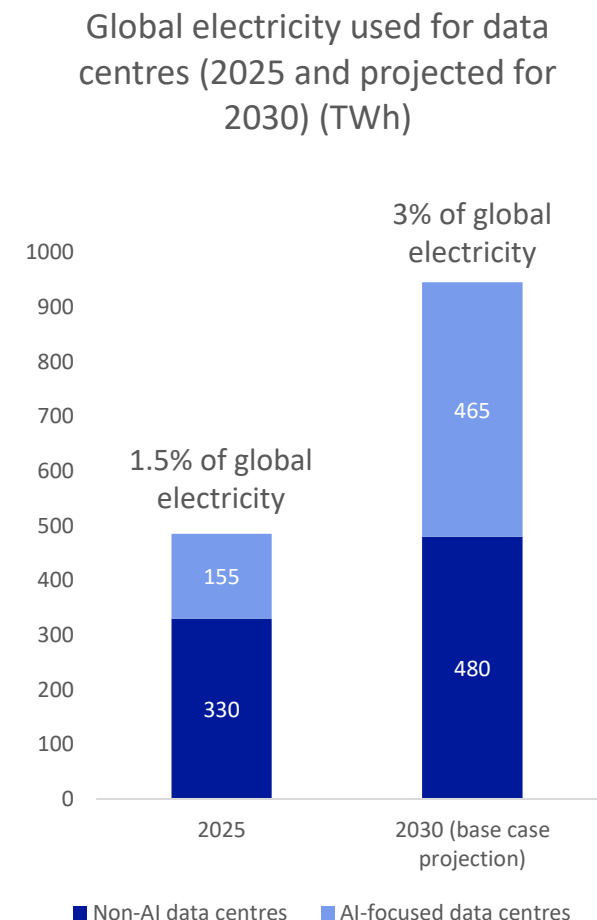
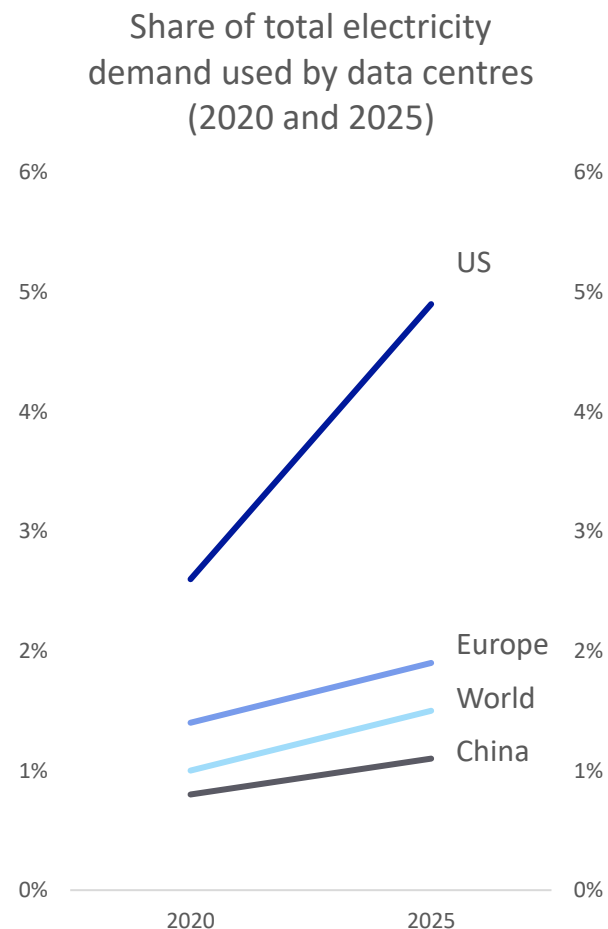
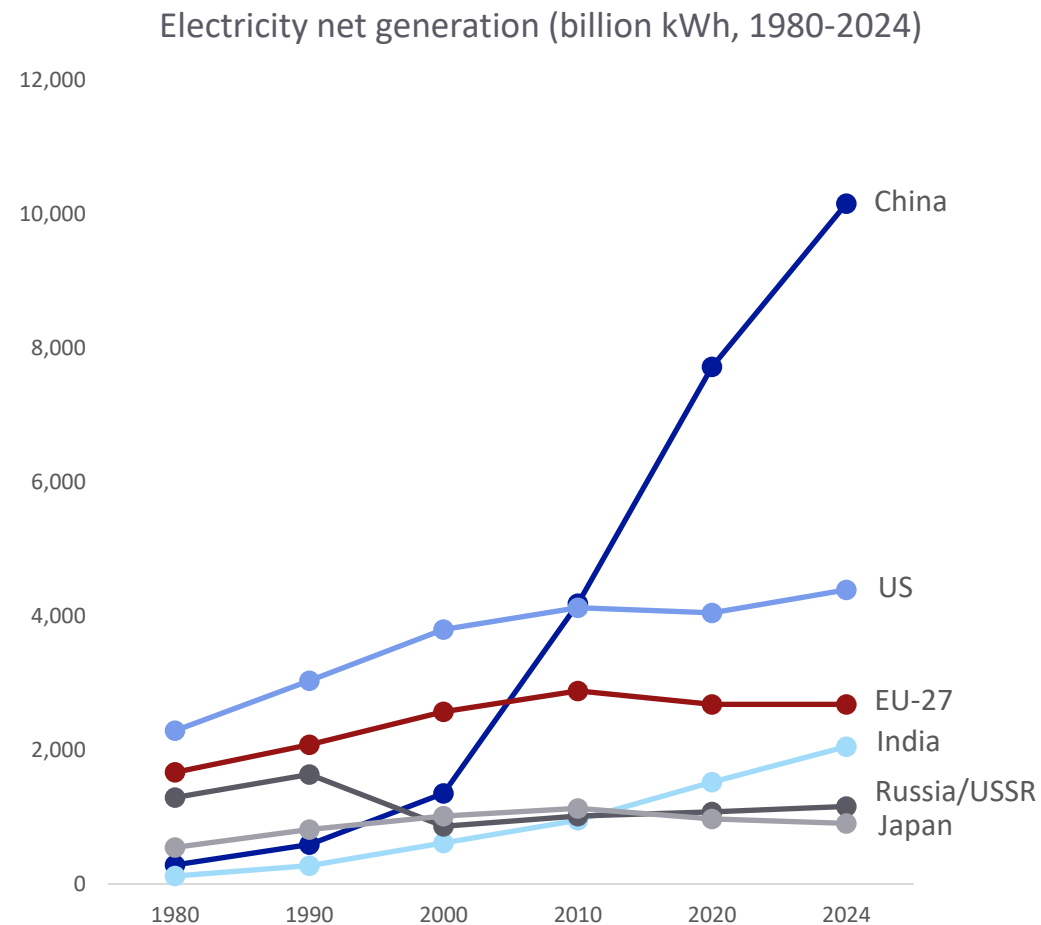
US enterprise AI adoption rate by two different measures (since Jan 2023)



Source: Our World in Data (citing IRENA, Nemet, Farmer and Lafond), Ramp AI Index, Deutsche Bank Research

6. China has left the US behind in electricity generation

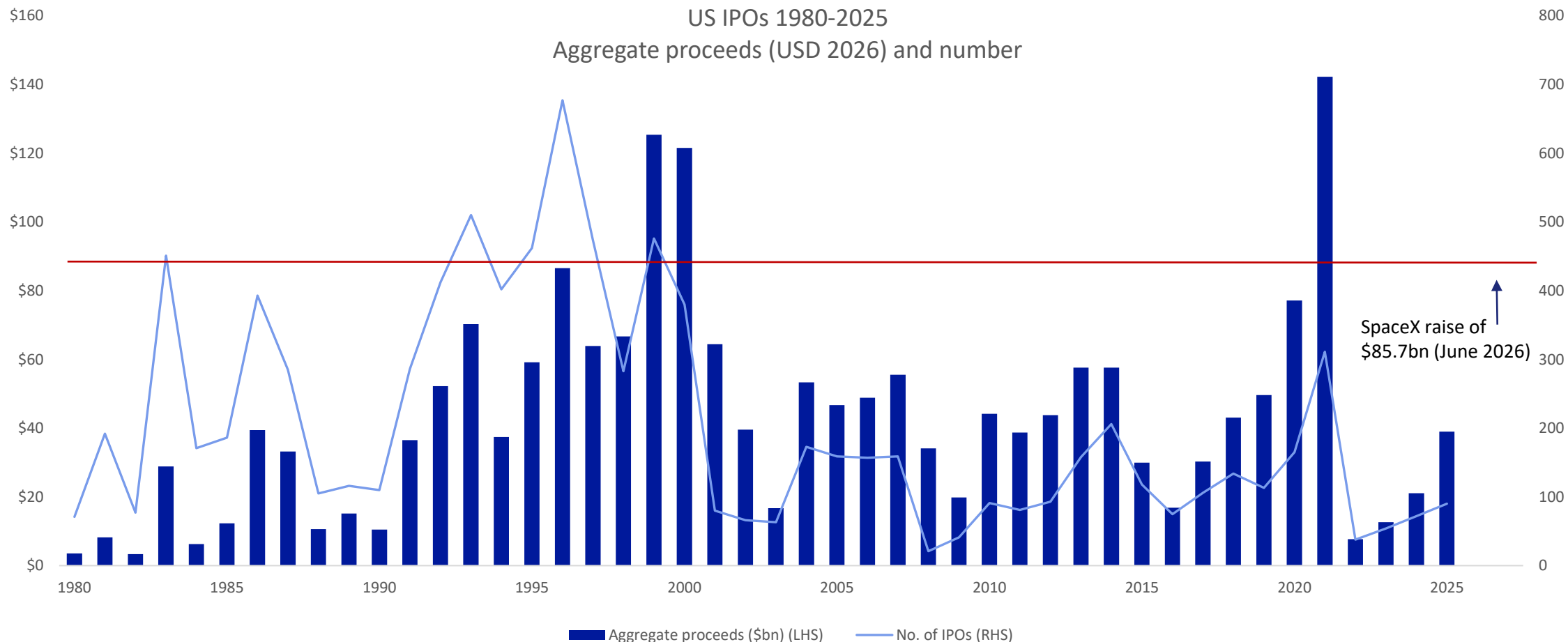
US is needing to use far more of its capacity for data centres even as demand continues to rise



Source: US Energy Information Administration, Our World in Data, International Energy Agency, Ember, Deutsche Bank Research

7. Mega AI IPOs will eclipse entire years of past share sales

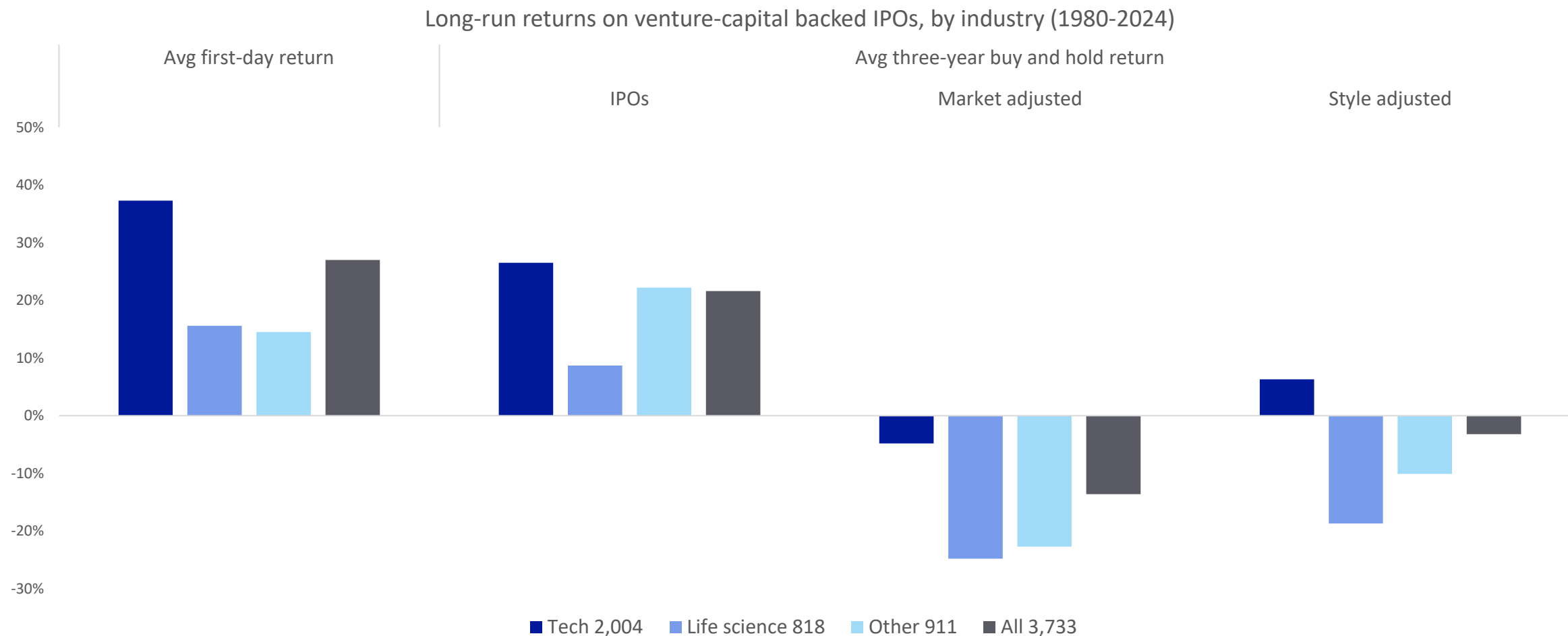
SpaceX's IPO exceeded all but three inflation-adjusted years



Source: Jay Ritter, University of Florida, Haver Analytics, Deutsche Bank Research

8. Tech IPOs have tended to outperform other IPOs

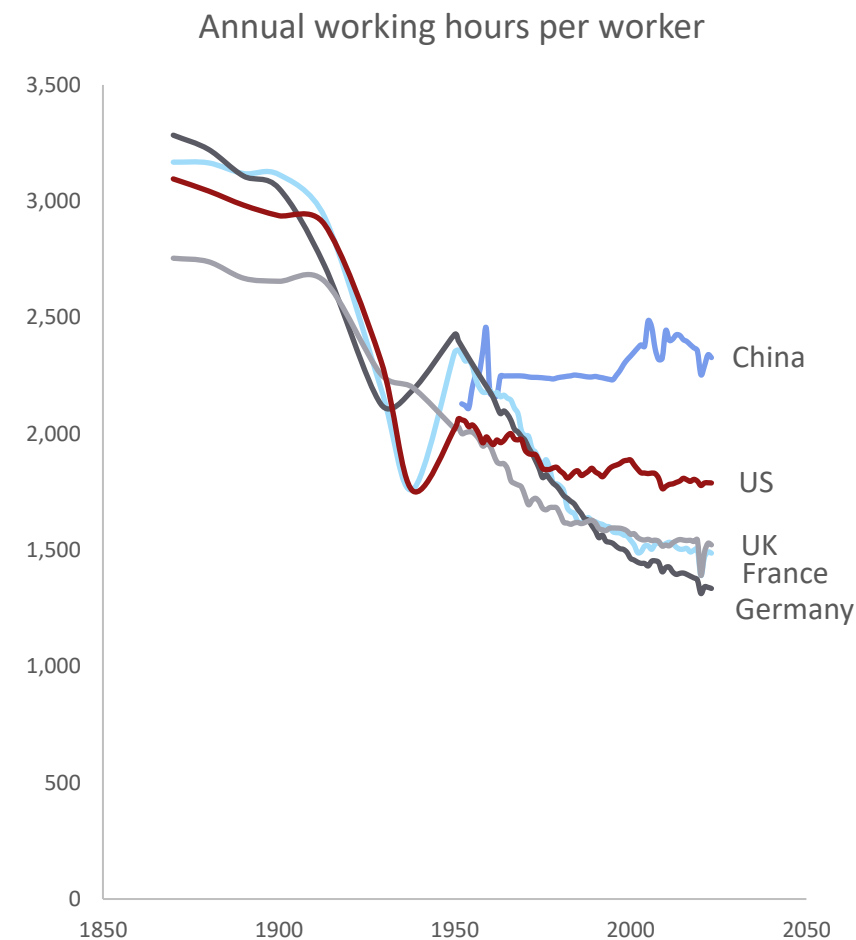
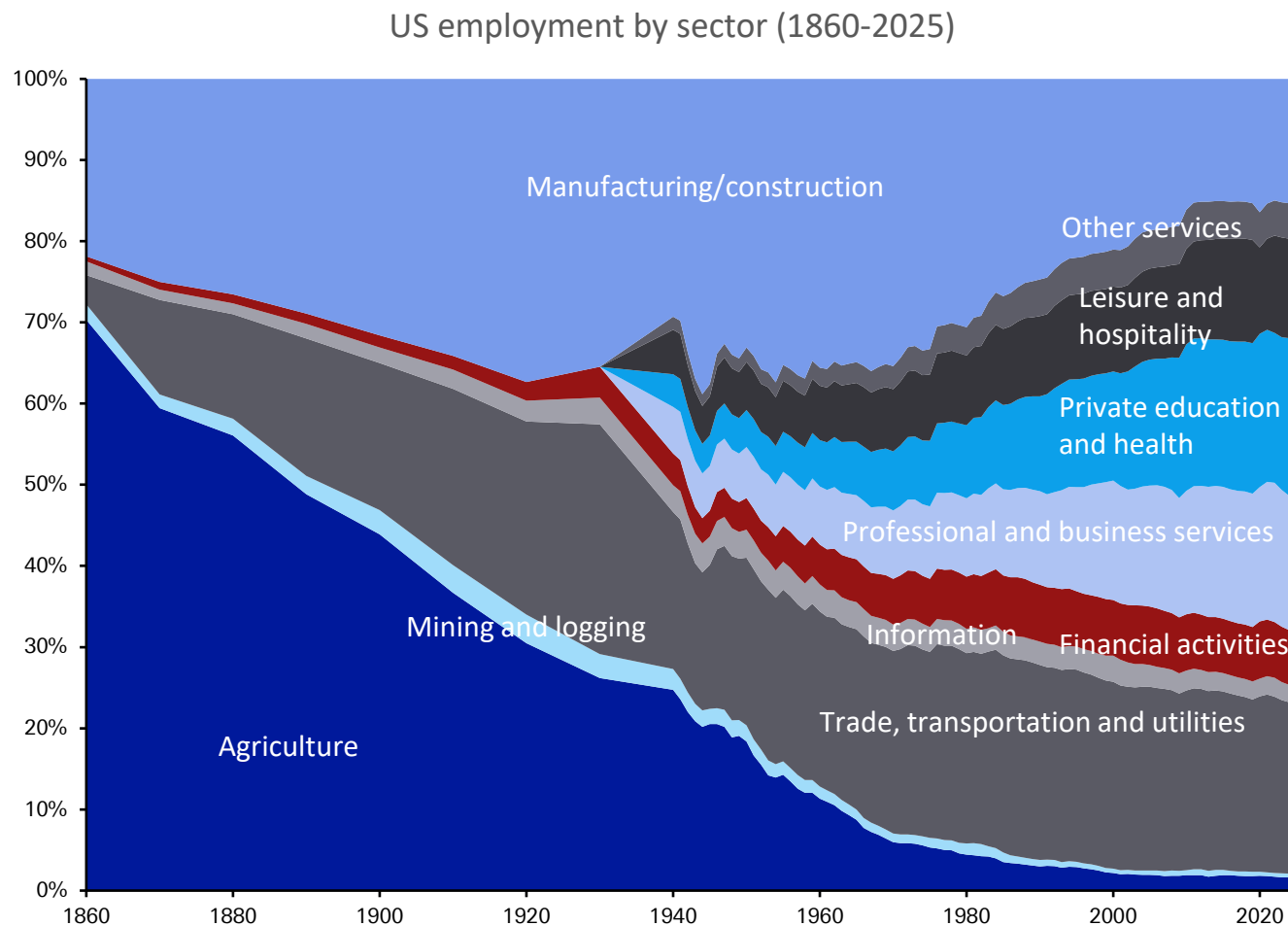
However, despite first-day surges, they have slightly underperformed the market over three years



Source: Jay Ritter, University of Florida, Haver Analytics, Deutsche Bank Research. Note: Three-year buy and hold calculated from first closing market price. Market-adjusted returns are the average buy-and-hold return minus the average compounded return on the Center for Research in Security Prices (CRSP) value-weighted index. Style adjustments are the equivalent for similar companies.

9. Jobs come and go but working hours have only fallen

As services have displaced agriculture then manufacturing, working hours have roughly halved



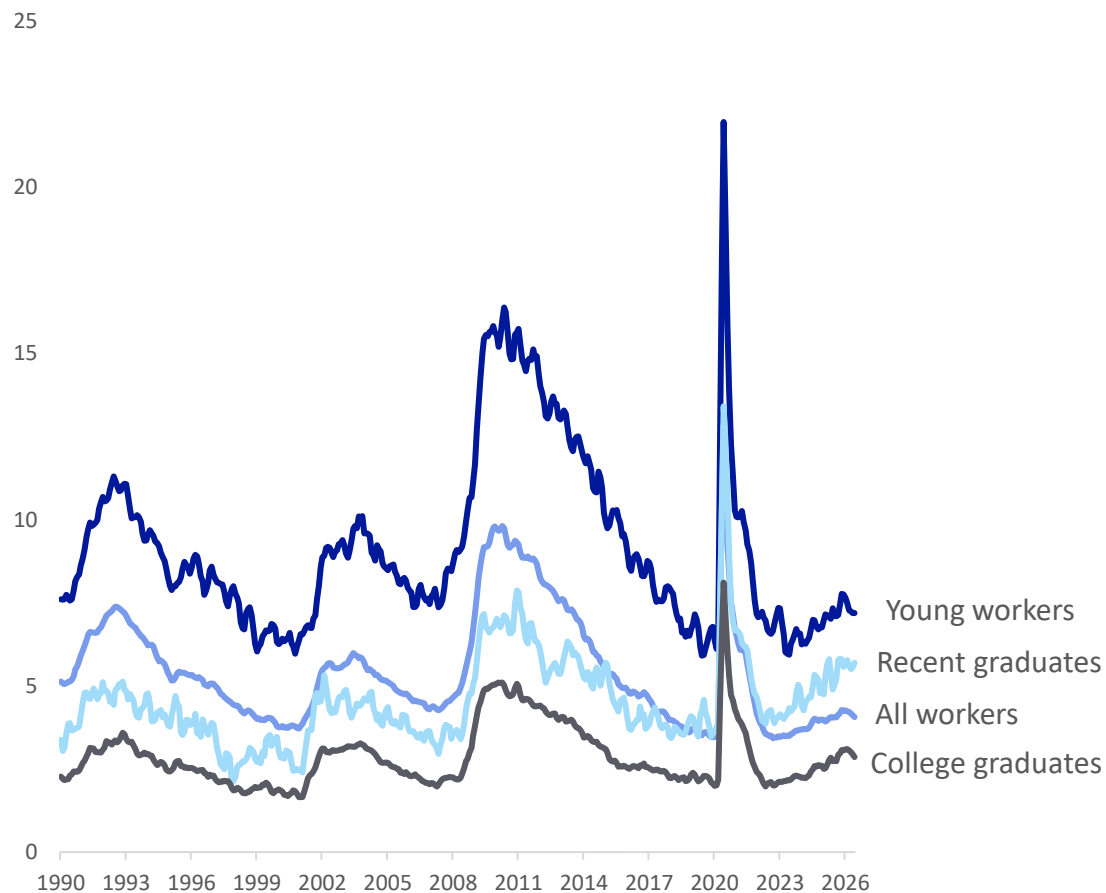
Source: Haver, Historical Statistics of the United States, 1789-1945, US Census, Our World in Data (citing Feenstra et al, Penn World Table (2025), Huberman and Minns (2005), Deutsche Bank Research

10. It's a tough job market for recent graduates

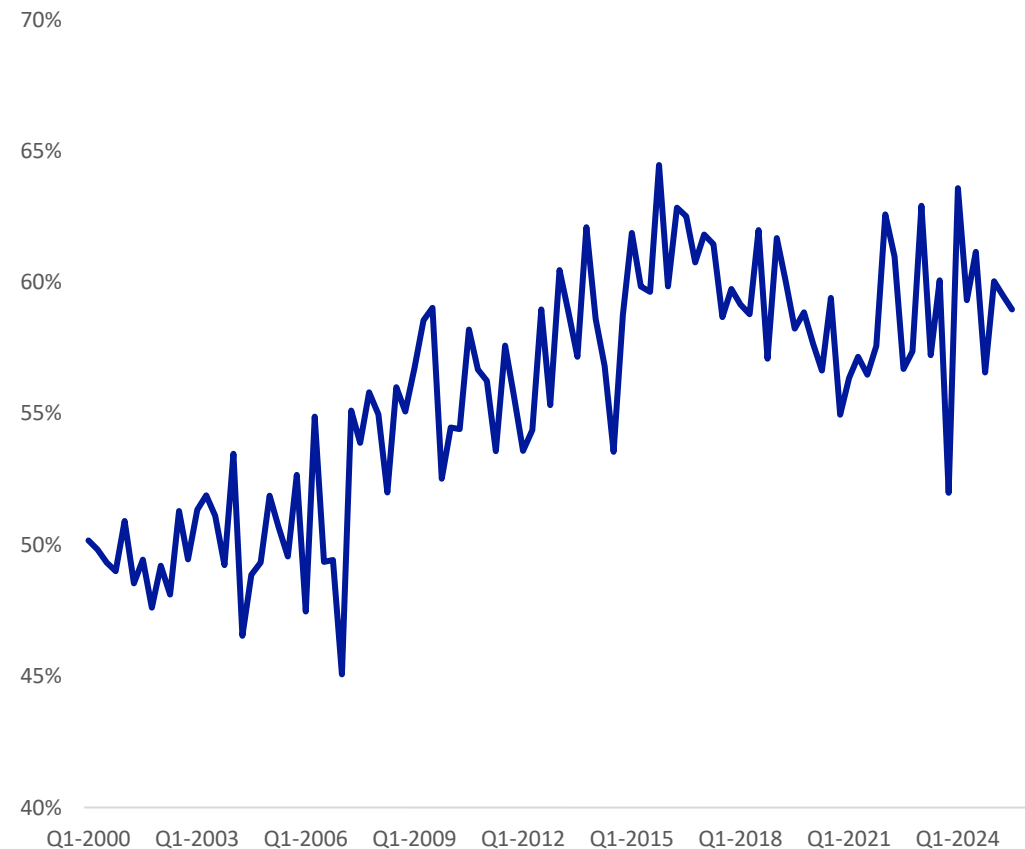
AI may be exacerbating the erosion over the past decade in graduate employment and wages



Unemployment rates for US workers



Bachelor's degree median weekly wage premium vs some college/associated degree (Q1 2000 to Q2 2026)



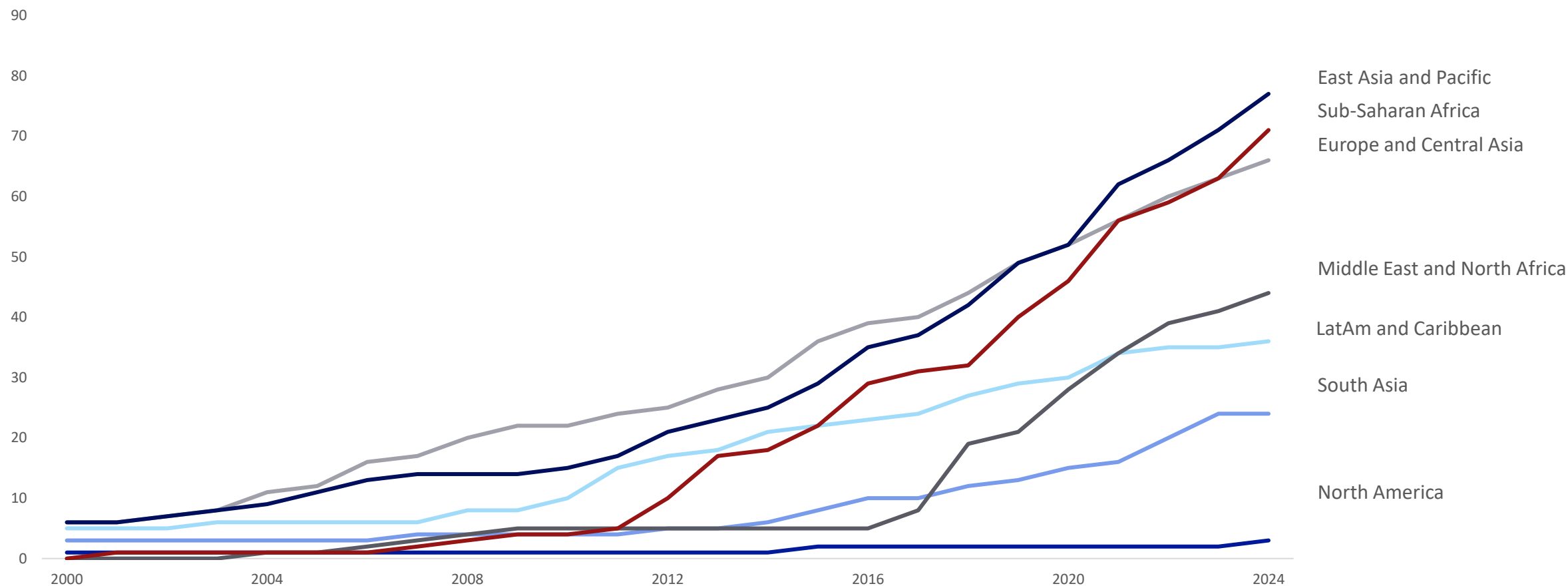
Source: U.S. Census Bureau, U.S. Bureau of Labor Statistics, Current Population Survey (IPUMS), Haver Analytics, Deutsche Bank Research. NB: For more, see Azar J, Mireia G and Sanz-Espin J, *The College Wage Premium in the Generative AI Era* (Aug 2026)

11. Concerns over data sovereignty date back long before AI

Countries have been restricting cross-border data flows, accelerating with EU's GDPR in 2016



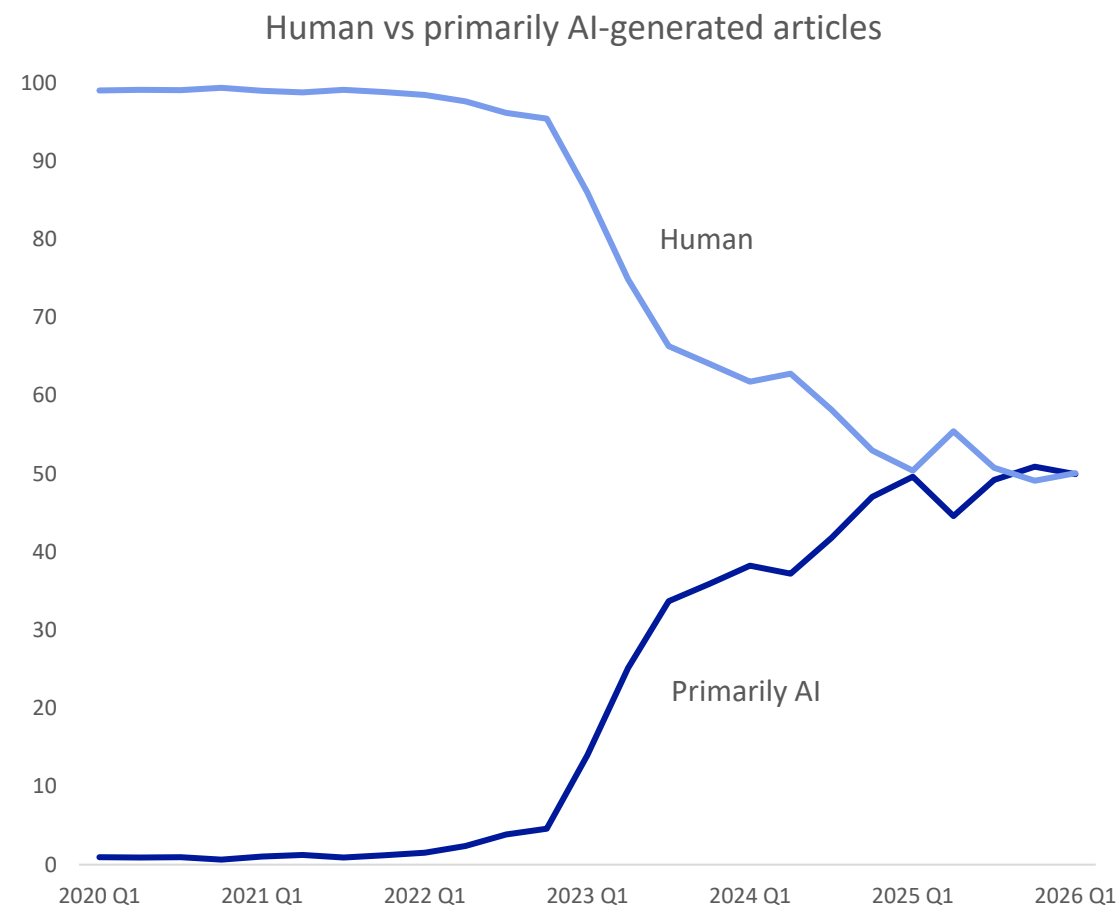
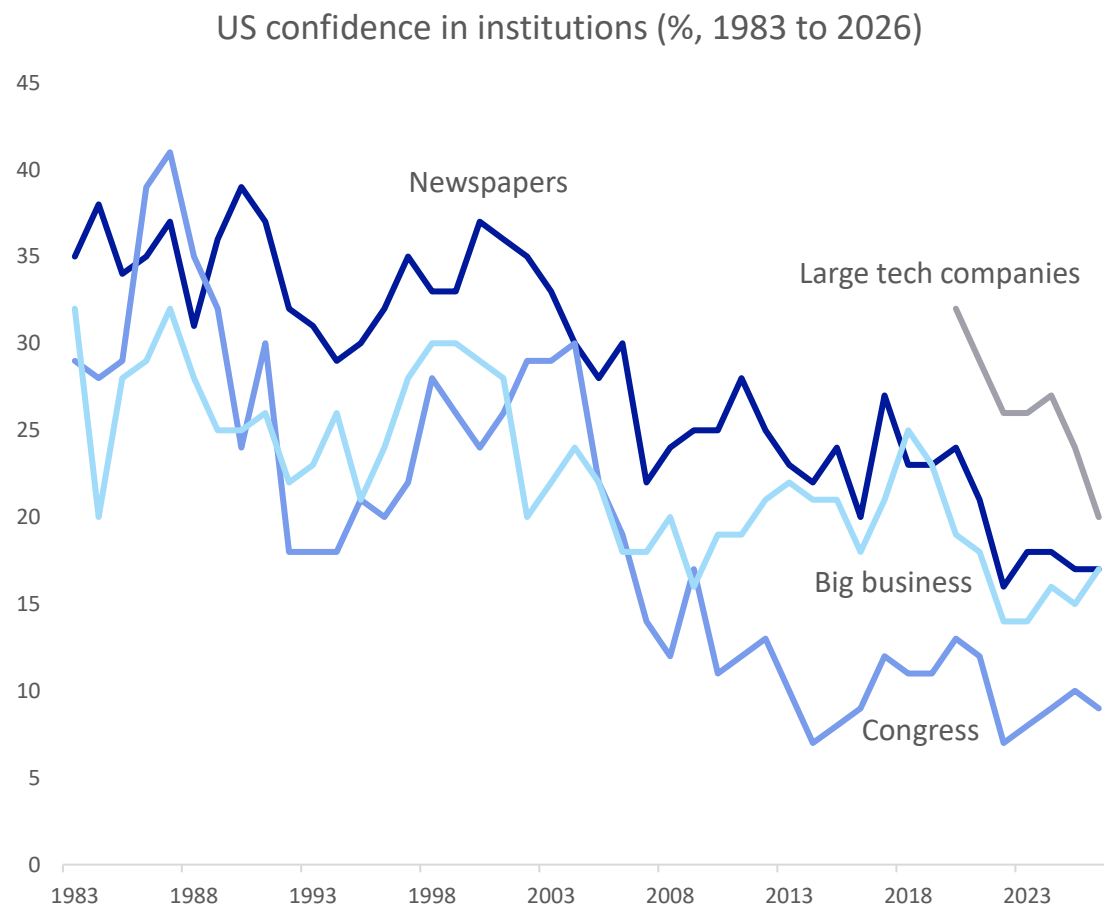
Data localisation measures by region (2000-2024)



Source: Stanford HAI AI Index (2026), Ferracane et al, Deutsche Bank Research

12. AI arrived as trust had been declining for decades

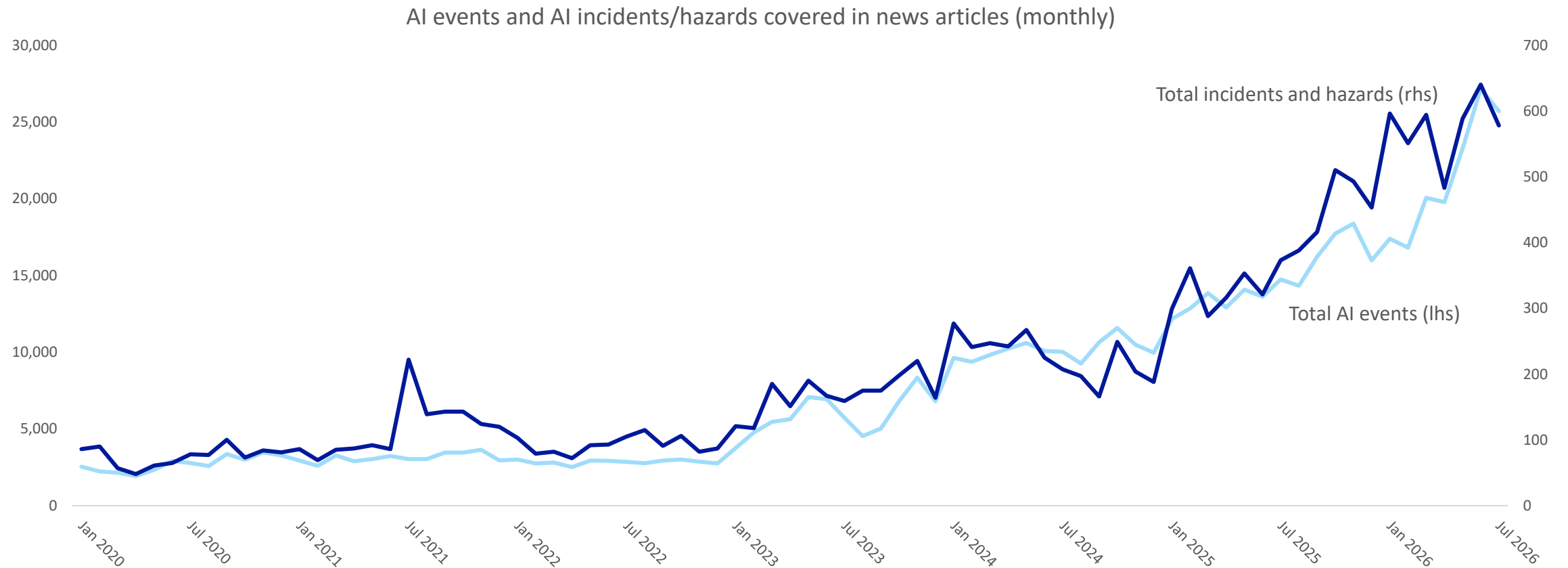
Confidence in institutions has plummeted – and now even human content is under threat



Source: Gallup ([Confidence in Institutions](#) | [Gallup Historical Trends](#)), [Graphite](#) (citing 55.4k articles from Common Crawl examined by AI detectors), Deutsche Bank Research

13. New technology brings new risks

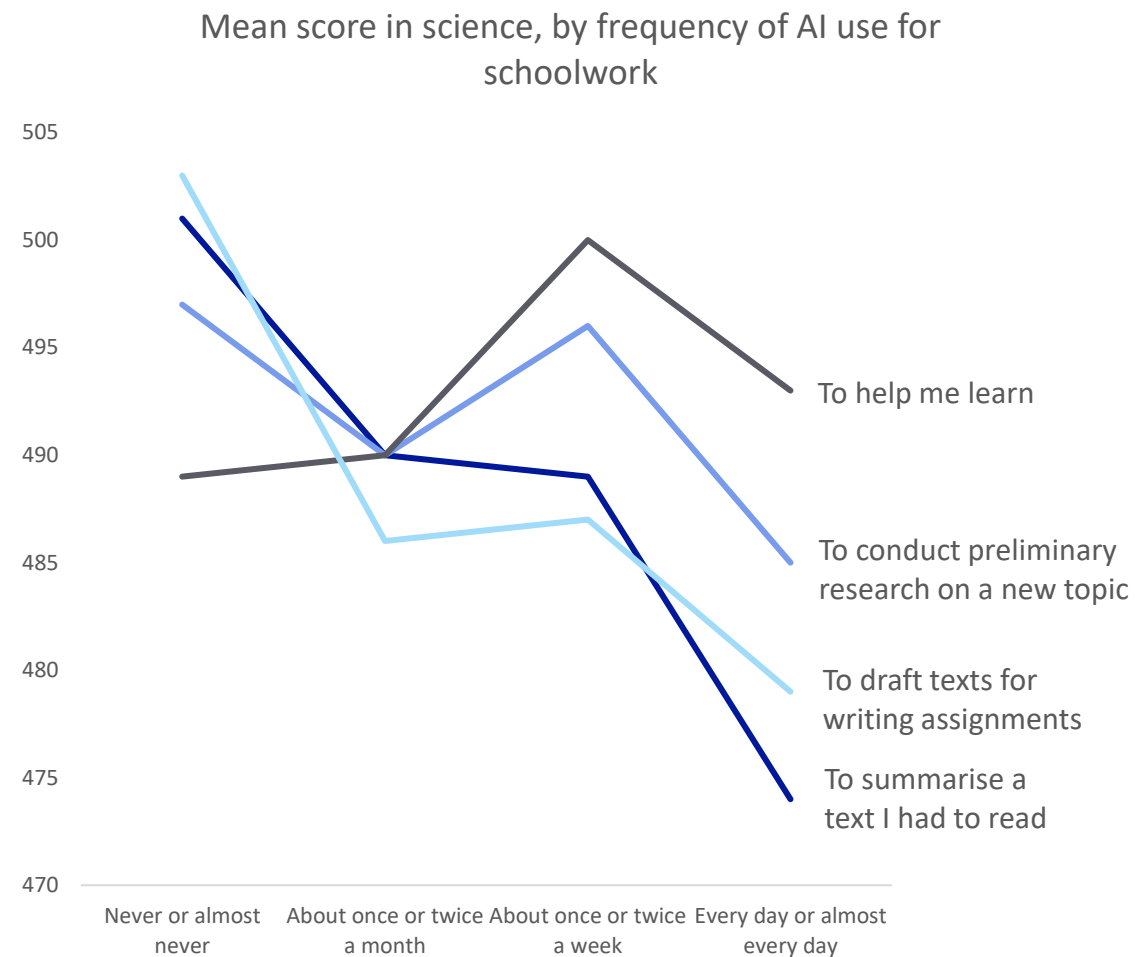
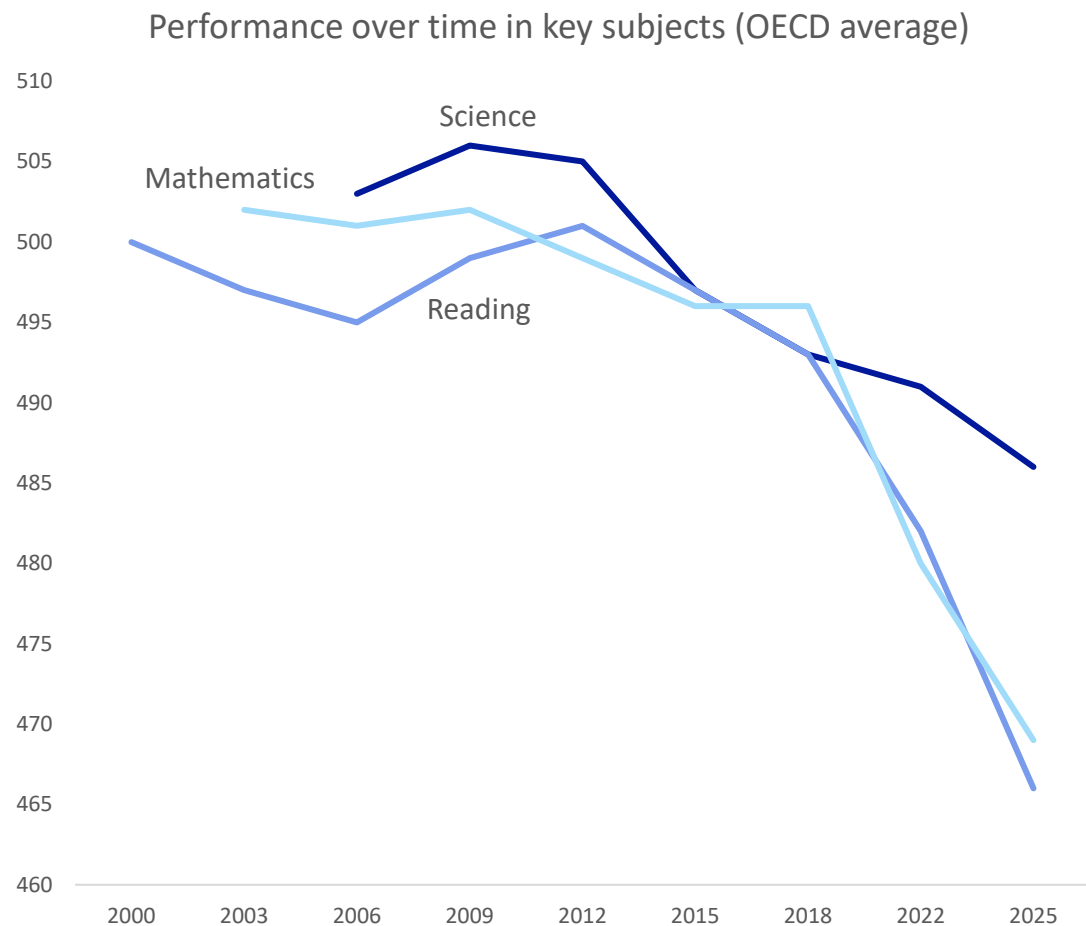
Coverage of AI events has surged, and so have hazards



Source: OECD AIM Incidents and Hazards Monitor, Deutsche Bank Research

14. Tech may have a role in declining educational performance

AI may be accelerating a fall that started with widespread smartphone diffusion from c. 2010



Source: OECD (2026), PISA 2025 Results (Volume I), Deutsche Bank Research



Appendix 1

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